Classifier model selection – Lesson overview

Video
Exercise
Examples

Understanding classifier model selection techniques is crucial for effectively working with different machine learning tasks. These techniques enable us to **evaluate** and **choose the most appropriate classifier algorithms** for a given dataset and problem domain.

In this lesson, we'll construct several **classification models**, conduct **cross-validation**, and **visualise** the **outcomes**. The focus will be on determining the most suitable model for a dataset based on **performance metrics**. Through hands-on exercises, we will gain valuable insights into model selection and validation processes.

Learning objectives

- Build multiple types of classification models, including logistic regression, k-nearest neighbours (KNN), support vector machines (SVM), decision trees and AdaBoost.
- Visualise the results of different classifiers.
- Perform cross-validation to assess the robustness of the models.